

Group Assignment

Instructions

1. This is a **group assignment**. Each group must consist of **eight (8) students**.
2. Each group must **select an existing or hypothetical company/organization** from any industry, such as healthcare, banking, education, manufacturing, agriculture, retail, logistics, hospitality, telecommunications, government, insurance, or any other approved sector.
3. Appropriate **diagrams, tables, Gantt charts, Work Breakdown Structures (WBS), organizational charts, risk matrices, flowcharts, and process models** should be included wherever applicable.

❖ Case Study

Your group has been contracted by a **client organization** to develop a **software system** that addresses a specific business need or operational challenge. The client organization may belong to any industry, such as healthcare, banking, education, manufacturing, agriculture, retail, logistics, hospitality, telecommunications, government, insurance, or any other sector approved by the lecturer.

The proposed software solution should be designed to improve the organization's operational efficiency, service delivery, decision-making, customer satisfaction, or overall business performance.

Each group is required to **select or create its own organization and software project**. The organization may be **real or hypothetical**, provided that sufficient background information and business context are developed.

For the purpose of this assignment, assume that the software project has the following characteristics:

- The project duration is **12 months**.
- The project is managed by a multidisciplinary software development team, which may include a Project Manager, Business Analyst, Software Developers, Software Testers, Database Administrator, UI/UX Designer, Quality Assurance Engineer, and other relevant personnel depending on the nature of the project.
- The project has defined stakeholders, business objectives, budget constraints, technical requirements, operational risks, and quality expectations.
- The software solution must comply with all applicable organizational policies, industry standards, legal requirements, and regulatory guidelines relevant to the selected organization.
- The software should demonstrate appropriate quality characteristics such as **functionality, reliability, security, usability, maintainability, scalability, availability, and performance**, according to the needs of the selected organization.
- Before software development begins, the client has requested the preparation of both a **Software Development Plan (SDP)** and a **Software Quality Assurance Plan (SQAP)** to ensure effective project management and quality assurance throughout the Software Development Life Cycle (SDLC).

As a software development consulting team, your responsibility is to analyze the selected organization, identify its business needs, propose an appropriate software solution, and prepare the required planning documents in accordance with Software Engineering and Software Quality Assurance best practices.

Questions

Question 1 (10 Marks)

Based on your selected organization and software project, you are required to:

- Describe the selected organization.
- Explain the organization's business activities.
- Identify the business problem or opportunity.
- Describe the proposed software solution.
- State the project objectives.
- Explain the expected benefits of the proposed software system.

Question 2 (15 Marks)

Develop a comprehensive **Software Development Plan (SDP)** for your proposed software project.

Your plan should include:

- Project objectives
- Project scope
- Development methodology
- Project organization
- Work Breakdown Structure (WBS)
- Project schedule
- Resource allocation
- Budget assumptions
- Risk management plan
- Communication plan

Question 3 (15 Marks)

Develop a comprehensive **Software Quality Assurance Plan (SQAP)** for the same project.

Your plan should include:

- Quality objectives
- Applicable standards and procedures
- Review and inspection activities
- Testing strategy
- Quality metrics
- Defect management process
- Quality audits
- Acceptance criteria
- Continuous improvement plan

Question 4 (10 Marks)

Compare the **Software Development Plan (SDP)** and the **Software Quality Assurance Plan (SQAP)** using a table containing **at least ten (10) comparison points**.

Question 5 (10 Marks)

Identify **at least five (5)** project risks that may affect your selected software project.

For each risk, you are required to:

- Describe the risk.
- Explain its possible impact on the project.
- Assess its likelihood or risk level.
- Recommend an appropriate mitigation strategy.

Present your answer in **tabular format**.

Question 6 (10 Marks)

Develop a **12-month Project Schedule** using either a **Gantt Chart** or a **project timeline** showing the major phases of software development.

Your schedule should include, but is not limited to:

- Project Initiation
- Requirements Analysis
- System Design
- Implementation
- Testing
- Deployment
- Maintenance

Question 7 (10 Marks)

Explain how the **Software Development Plan** and the **Software Quality Assurance Plan** work together throughout the **Software Development Life Cycle (SDLC)**.

Support your explanation with an appropriate **flowchart, workflow diagram, or process model**.

Question 8 (10 Marks)

Assume that your software project changes from a **12-month external client project** to a **2-month internal organizational project**.

Discuss:

- Which components of the Software Development Plan would be simplified?
- Which Software Quality Assurance activities would remain essential?

- Why project planning and documentation should still be maintained despite the reduced project size.

Question 9 (10 Marks)

Discuss the possible consequences if the selected organization decides to begin software development **without preparing a Software Development Plan (SDP) and a Software Quality Assurance Plan (SQAP)**.

Support your discussion with practical software engineering examples relevant to your selected project.

END.